

SOA Survey

Contextual Sign Disambiguation: Moving from Sign Classification to Automated Sentence Translation

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Abstract

This survey investigates the evolution of Ancient Egyptian hieroglyph recognition from plain sign classification toward contextual sentence translation. While isolated symbol identification has achieved high accuracy, the field currently lacks an automated mechanism to “read” entire documents where meaning is dependent on neighboring context and “determinatives.” We reviewed 10 papers spanning instance segmentation, custom CNN architectures (e.g., Glyphnet), and multi-modal script homogenization. We identified a critical research gap: existing models treat hieroglyphs as independent objects, failing to resolve visual ambiguities or generalize across different scribal hands and archaeological supports. To address this, we propose the implementation of Attention-based Vision-Language Transformers. By employing Multi-Head Attention, we plan to move beyond the current limitation of sign-by-sign identification toward holistic contextual understanding. This direction leverages “Contextual Sign Disambiguation” to correct visual misclassifications, such as confusing nearly identical bird signs, by using linguistic context to infer meaning. This report outlines the technical requirements and data landscape necessary to enable end-to-end decipherment from archaeological imagery within a modern deep learning framework.

This matters because thousands of artifacts remain untranslated due to the global shortage of Egyptological experts. Automated translation would provide a bridge between raw visual data and historical interpretation. From a deep learning standpoint, this is hard because of the “**long-tail**” **distribution of data** (700+ signs with very few samples for many) and **domain shift** (carved stone looks different than ink on papyrus).

Furthermore, current systems are “context-blind.” They classify a sign based solely on its pixels, whereas the language requires “Contextual Sign Disambiguation.” Resolving this requires an architecture capable of modeling long-range dependencies between symbols, which is why a transition to Attention-based models is scientifically justified.

1. Problem Definition

The primary task is the transition from individual hieroglyphic sign identification to the automated translation of complete sentences. Current pipelines are effective at **Segmentation** (outlining signs) [1] and **Classification** (naming signs like “G17”) [2], but they fail to “read” the language. In Egyptian hieroglyphs, a sign’s identity is often fluid [1]; it can be a phonetic sound, an entire word, or a “determinative” that defines the category of the previous word. Without understanding the sequence, a model cannot accurately transliterate or translate the text.

2. Literature Review**2.1. Challenge 1: Isolated Sign Identification**

The baseline for this field was established by papers focusing on naming individual signs. Barucci et al. (2021) proposed “Glyphnet,” a custom CNN architecture tailored for Egyptian ideograms [2]. They proved that specialized models outperform general-purpose networks like ResNet-50. However, the study was limited to 40 classes and ignored the sequence in which signs appear [2]. Because the model only looks at one sign at a time, it cannot fix mistakes where two signs are visually similar but linguistically distinct.

2.2. Challenge 2: Multi-Modal and Scribal Variability

Hieroglyphs appear on stone, wood, and papyrus, and vary significantly based on the “hand” of the scribe [1, 4]. Wang et al. (2025) addressed this using U-Net based homogenization to unify rubbings and drawings. This was a critical step in handling “domain shift.” [4] Yet, even with homogenized data, models struggle to identify whether a variation in a sign is a different character or simply a stylistic choice by a specific scribe. Scribe identification remains an open problem that requires deeper feature extraction than current CNNs provide.

2.3. Challenge 3: The Failure of Shallow Context

Early attempts at contextual recognition, such as Franken et al. (2013), used N-grams to re-rank visual guesses [3]. N-grams only look at immediate neighbors (e.g., 2 or 3 signs). Since hieroglyphs often rely on determinatives located at the end of a word to clarify symbols at the beginning, these shallow models fall short. This highlights the need for **Attention models**, which can create relations between distant words to derive better understanding of the sentence as a whole.

Open Problems (flagged by the field:)

- **Data Scarcity for Attention:** Transformer models need extensive datasets to learn the logic of a language, but hieroglyphic data is scarce [3].
- **Out-of-set symbols:** Encountering a sign not in the training set leads to silent misclassification [2].
- **Reading Order/Toposyntax:** Hieroglyphs are stacked in 2D blocks; determining the correct reading sequence automatically is still unsolved [3, ?].

3. Research Gap and Proposed Direction

The clear research gap is the disconnect between **visual identification** and **linguistic understanding**. Prior work has successfully named signs but struggles to interpret them in sequence. There is no existing framework that uses **Attention** to resolve visual ambiguities (like “Bird-type” errors [3]) through long-range linguistic context. Additionally, current models

are unable to distinguish between signs and “scribal hand” variations.

We therefore propose to investigate a **Vision-Language Transformer** approach. Prior work does plain classification; we therefore propose to investigate **Contextual Sign Disambiguation via Attention**. By treating a row of hieroglyphs as a sequence rather than a bag of objects, we can leverage neighboring symbols to infer the identity and meaning of ambiguous signs.

Research Question: Can a Multi-Head Attention mechanism significantly improve the recognition of ambiguous hieroglyphs by leveraging long-range linguistic context, and can it effectively normalize variations across different scribal hands?

4. Benchmark, Evaluation, and Data Landscape

Datasets. Key datasets identified in the surveyed literature.

Table 1: Key datasets in the field.

Dataset	Size	Modality	Notes
OBC306	165 cls	Rubbings	[3, 4]
HWOB	83k img	Handdrawn	[4]
Barucci	4.3k	Mixed	[2]
D1/D2	img		
Fabricius	Large	Photo	[GitHub Link]
EHT Dataset	4k+ seq	Text	[GitHub Link]

Evaluation Protocol. The field uses **Top-1 Accuracy** for naming. For translation, we will adopt **BLEU Scores** and **WER** (Word Error Rate) to measure the accuracy of the English sentence output.

Compute Context. Surveyed papers primarily used NVIDIA Tesla T4 or RTX 4090 GPUs [1, 2]. Transformer training will require 24GB+ VRAM to handle attention over full-image patches [5, 7].

Experimental Design Considerations

- **Baselines:** We will compare against **Glyphnet** (classification baseline) [2] and the **N-gram pipeline** (translation baseline) [3].
- **Ablation Study:** We will isolate the **Attention layers** [5] to quantify exactly how much contextual information fixes isolated sign misclassifications.
- **Pitfalls:** Data Leakage (training/testing on the same stela) and Dataset Bias (models favoring carved stone over papyrus).

5. Annotated Reading List

References

- [1] A. Barucci, et al. Segmentation of ancient Egyptian hieroglyphs with Detectron2. *Algorithms*, 16(2), 79, 2023. <https://doi.org/10.3390/a16020079> *What it does:* Uses Mask R-CNN to segment symbols on various supports. *What it contributes:* A robust dataset creation and segmentation pipeline. *Why it is here:* Sets the baseline for identifying symbols in noisy images.
- [2] A. Barucci, et al. Deep Learning Approach to Ancient Egyptian Hieroglyphs Classification. *IEEE Access*, 9, 2021. <https://ieeexplore.ieee.org/document/9528382> *What it does:* Proposes "Glyphnet" for 40-class sign identification. *What it contributes:* Proves specialized CNNs beat general ones like ResNet. *Why it is here:* Our primary classification baseline.
- [3] M. Franken, et al. Automatic Recognition of Ancient Egyptian Hieroglyphs. *ACM Multimedia*, 2013. <https://doi.org/10.1145/2502081.2502199> *What it does:* Full pipeline using visual matching and N-grams. *What it contributes:* First attempt at using linguistic context for hieroglyphs. *Why it is here:* Highlights the limitations of shallow language models.
- [4] N. Wang, et al. Multi-modal ancient scripts recognition via deep learning. *npj Heritage Science*, 13(522), 2025. <https://www.nature.com/articles/s40494-025-02095-x> *What it does:* Uses U-Net for data homogenization across rubbings. *What it contributes:* Framework for handling scribal hand variations. *Why it is here:* Addresses the material and handwriting shift problem.
- [5] A. Vaswani, et al. Attention is All You Need. *NeurIPS*, 2017. <https://doi.org/10.48550/arXiv.1706.03762> *What it does:* Introduces the Transformer and Self-Attention. *What it contributes:* The mechanism for modeling long-range dependencies. *Why it is here:* Technical foundation for our contextual translation approach.
- [6] X. Li, et al. TrOCR: Transformer-based Optical Character Recognition. *AAAI*, 2023. <https://doi.org/10.48550/arXiv.2109.10282> *What it does:* End-to-end image-to-text transformer encoder-decoder. *What it contributes:* Bypasses separate segmentation stages for sequence text. *Why it is here:* Reference for end-to-end translation goal.
- [7] A. Dosovitskiy, et al. An Image is Worth 16x16 Words. *ICLR*, 2021. <https://doi.org/10.48550/arXiv.2010.11929> *What it does:* Applies Transformers directly to image patches (ViT). *What it contributes:* Replaces traditional CNNs with Self-Attention. *Why it is here:* Tech to be used for the vision backbone.
- [8] A. Radford, et al. Learning Transferable Visual Models from Natural Language. *ICML*, 2021. <https://doi.org/10.48550/arXiv.2103.00020> *What it does:* Connects images to text descriptions in a shared space. *What it contributes:* Enables Few-shot/Zero-shot recognition of rare signs. *Why it is here:* Directly addresses the "Out-of-set symbols" challenge.
- [9] H. Chefer, et al. Transformer Interpretability Beyond Attention Visualization. *CVPR*, 2021. https://openaccess.thecvf.com/content/CVPR2021/html/Chefer_Transformer_Interpretability_

- [Beyond_Attention_Visualization_CVPR_2021_paper.html](#) What it does: Methods to visualize exactly what pixels a Transformer sees. What it contributes: Extracts heatmaps showing which pixels triggered a decision. Why it is here: Used to identify stylistic "Scribe signatures" through attention.
- [10] Y. Huang, et al. LayoutLMv3: Pre-training for Document AI. *ACM Multimedia*, 2022. <https://dl.acm.org/doi/10.1145/3503161.3548112> What it does: Models text, position, and image layout simultaneously. What it contributes: Handles complex 2D document structures (toposyntax). Why it is here: Essential for solving the reading order and stacked sign problem.
- [11] R. Fuentes-Ferrer, et al. Recognition of Egyptian hieroglyphic texts through focused generic segmentation and cross-validation voting. *Applied Soft Computing*, 112793, 2025. <https://doi.org/10.1016/j.asoc.2025.112793> What it does: Employs the Segment Anything Model (SAM) for zero-shot segmentation combined with a ConvNeXt classification backbone and an ensemble voting mechanism. What it contributes: Introduces a highly scalable end-to-end pipeline that achieves human-level recognition accuracy across 300+ hieroglyph categories on heterogeneous archaeological datasets. Why it is here: It represents the 2025 state-of-the-art for high-recall recognition systems, serving as the definitive visual performance baseline for our proposed research.
- [12] S. F. Ferrer, et al. Neural Style Transfer for Synthesising a Dataset of Ancient Egyptian Hieroglyphs. *arXiv preprint*, 2024. <https://arxiv.org/abs/2504.02163> What it does: Employs Neural Style Transfer to apply stone textures to digital font shapes. What it contributes: Provides a scalable solution to the data scarcity and domain shift problem. Why it is here: Justifies our use of synthetic data to train contextual attention models.
- [13] S. El-Taher, et al. HieroLM: Egyptian Hieroglyph Recovery with Next Word Prediction Language Model. *ACL Anthology*, 2024. <https://arxiv.org/abs/2503.04996> What it does: Implements a linguistic recovery model using LSTMs for missing symbol prediction. What it contributes: Shifts focus from visual recognition to statistical linguistic inference. Why it is here: Acts as the primary baseline for our contextual disambiguation research.
- [14] C. Cucci, et al. AI-based methods and hyperspectral imaging for automated translation. *SPIE Cultural Heritage*, 2025. <https://doi.org/10.1117/12.3066645> What it does: Uses hyperspectral 3D data cubes to see pigments invisible to RGB cameras. What it contributes: Recovers legibility on severely degraded surfaces using physics-informed AI. Why it is here: Highlights the open problem of data acquisition for corrupted artifacts.

6. Preliminary Implementation Resources

- <https://github.com/facebookresearch/detectron2>
Implementation of Mask R-CNN used as the primary baseline for symbol segmentation tasks [1].
- <https://github.com/GAIA-IFAC-CNR/Glyphnet>
Official repository for the Glyphnet architecture; serves as our primary baseline for sign classification [2].
- <https://github.com/morrisfranken/glyphreader>
Source for the Franken (2013) pipeline and access point for the Pyramid of Unas (OBC306) dataset [3].
- <https://github.com/googlearts/culture/workbench>
Primary dataset resource and implementation logic for Google's Fabricius experiment.
- <https://github.com/rfuentesfe/EgyptianHieroglyphicText/tree/main/dataset>
Linguistic dataset containing sequential Gardiner

codes and text-based Egyptian corpora.

- <https://github.com/microsoft/unilm/tree/master/trocr>
Core backbone for transformer-based sequence recognition and end-to-end character transcription [6].
- https://github.com/dqj5182/Weighted_Random_Sampling_for_Egyptian_Hieroglyph_Classification
Specific utilities for handling the long-tail distribution and class imbalance in hieroglyphic datasets.
- <https://github.com/Augety88/oracle-jinwen-code-data>
Reference implementation for multi-modal script homogenization and generative data augmentation [4].
- <https://github.com/rfuentesfe/EgyptianHieroglyphicText>
Linguistic corpus of sequential Gardiner codes used for training contextual attention models.